

EXPERIMENT - 5

TO DETERMINE THE EQUIVALENT WEIGHT OF GIVEN METAL BY HYDROGEN DISPLACEMENT METHOD.

APPARATUS REQUIRED

- | | |
|-----------------------------|----------------------------|
| a) <u>Eudiometer</u> | b) <u>Beaker</u> |
| c) <u>Short stem funnel</u> | d) <u>Clamp with stand</u> |
| e) <u>Tall jar</u> | f) <u>Thermometer</u> |
| g) <u>Chemical balance</u> | h) _____ |

CHEMICAL REQUIRED

- | | |
|-------------------------------------|---------------------------------|
| i. <u>Dil. H_2SO_4</u> | ii. <u>A piece of Mg ribbon</u> |
|-------------------------------------|---------------------------------|

THEORY

Equivalent weight of an element is defined as the number of parts by weight of the element that can combine with or displace from a compound 1.008 parts by weight of hydrogen or 8 parts by weight of oxygen or 35.5 parts by weight of chlorine.

Hydrogen displacement method is used to determine the equivalent weight of those metals which are more electropositive than Hydrogen because the metals like Zn, Al, Mg etc can displace hydrogen from dilute mineral acid like HCl, H_2SO_4 etc.

In this method, A known weight of metal is reacted with excess dilute sulphuric

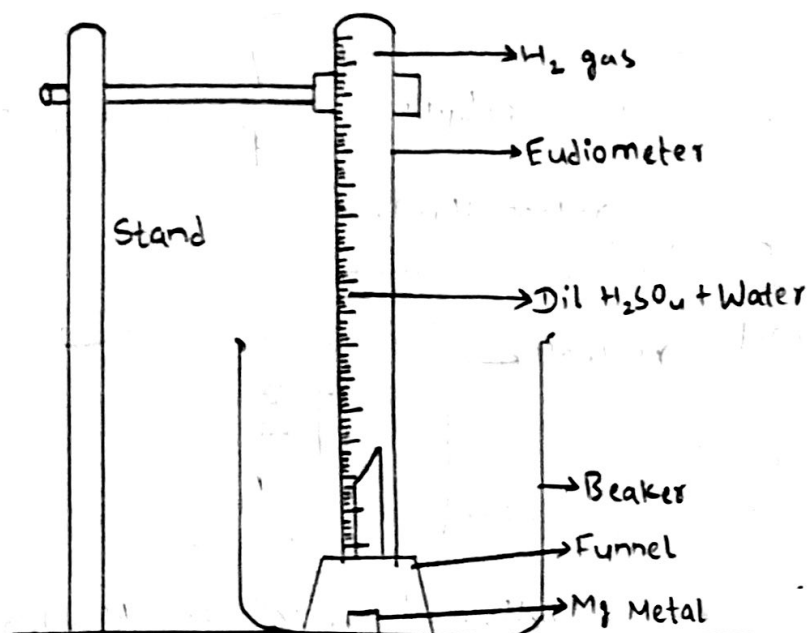
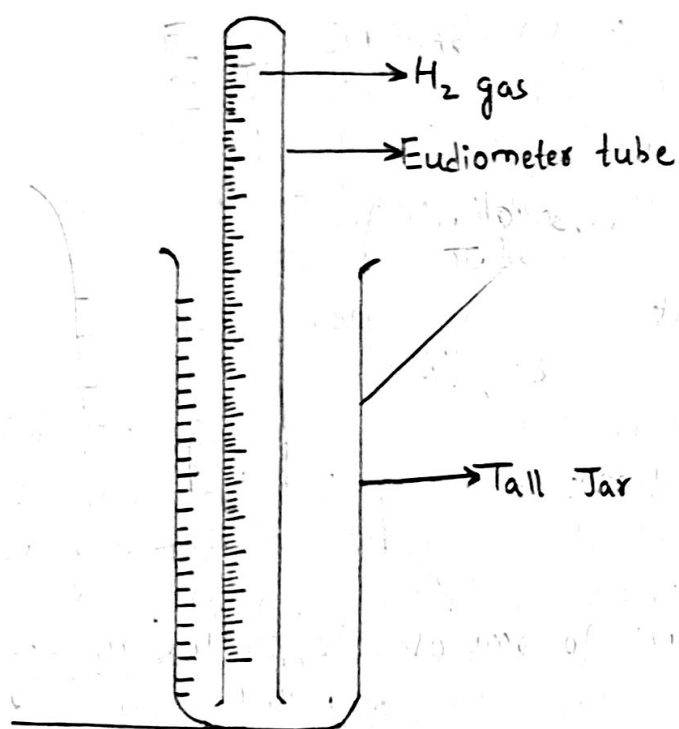


fig - Hydrogen displacement method

fig - Calibration of Volume of H_2 gas
using Tall Jar

acid or dilute hydrochloric acid



The volume of H_2 gas evolved is measured. The volume of moist hydrogen gas collected in the tube is converted in dry volume at NTP by applying combined gas equation and Dalton's law of partial pressure. The volume at NTP is then converted into its weight and the equivalent weight of metal is determined by using following Formula.

$$\text{Equivalent Weight of metal} = \frac{\text{Weight of metal}}{\text{Weight of H}_2 \text{ displaced}} \times 1.008$$

PROCEDURE

A known weight of metal was taken in a beaker and was covered with a short-stem funnel. Water was added slowly to the beaker so that the stem of the funnel was kept completely immersed in the water. An eudiometer tube was filled more than half with dilute H_2SO_4 and the remaining portion was filled with water. Its mouth was closed with the thumb and it was carefully inverted over the stem of the funnel so that no air bubbles were allowed to enter the tube. As soon as the metal came in contact with the acid, a reaction was produced and hydrogen gas was evolved. After the reaction was completed, the mouth of the eudiometer tube was closed under water and the tube was inserted into a tall

jar containing water in an inverted position. The water levels inside and outside the tube were adjusted to be equal, and the volume of moist gas at the laboratory temperature was measured. The atmospheric pressure, laboratory temperature, and aqueous tension were also recorded.

OBSERVATION

Mass of metal = 0.03 gm

Volume of hydrogen gas collected = 35.5 ml

Lab temperature = 21°C

Atm. pressure = 650

Aqueous tension at $t^{\circ}\text{C}$ = 18.7

pressure exerted by dry gas = $(P - f)$ mm of Hg
 $= 631.5$ mm of Hg

CALCULATION

Weight of metal taken = 0.03 gm

At Laboratory temp. & pressure

$V_1 = 35.5$ ml

$P_1 = 650$ mm Hg

$T_1 = 294$ K

At Normal Temperature pressure (NTP)

$V_2 = ?$

$P_2 = 760$ mm of Hg

$T_2 = 273$ K

From combined gas equation

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\text{or, } V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$= \frac{631.5 \times 35.5 \times 273}{760 \times 294} = 28.19$$

$$\text{Weight of dry H}_2 = 27.38 \times 0.000089 = 2.43 \times 10^{-3} \text{ gm}$$

$$\text{Now, Equivalent weight} = \frac{\text{Weight of Metal}}{\text{Weight of H}_2} \times 1.008$$

$$= \frac{0.03}{2.43 \times 10^{-3}} \times 1.008$$

$$= 12.05$$

RESULT

In this way the equivalent weight of the given metal was determined by Hydrogen displacement method and is found to be 12.43 gram.

CONCLUSION

Hence, From above exp. it can be concluded that the equivalent weight of metals which are electro-positive than hydrogen can be determined by hydrogen displacement method.

PRECAUTIONS

1. Personal protective equipment should worn.
2. Reaction between metal and acid should be completed.
3. There should not be any leakage of hydrogen gas.
4. Eudiometer containing dil. H_2SO_4 should be free of air bubbles.
5. In the tall jar, outer and inner levels should be coincided accurately to get accurate volume of H_2 gas.

VIVA-VOCE

1. Can we find the equivalent weight of Cu metal by hydrogen displacement method?
2. Why is volume of hydrogen gas collected in calibrated eudiometer tube by placing it inside tall jar containing water?
3. How Dalton's law of partial pressure is applied in this experiment?
4. What is equivalent weight of metal?
5. What is equivalent weight of H_3PO_4 in following equation

$$3\text{Mg} + 2\text{H}_3\text{PO}_4 \longrightarrow \text{Mg}_3(\text{PO}_4)_2 + 3\text{H}_2$$

1. No, we cannot because copper metal lies below the hydrogen in electrochemical series which cannot displace hydrogen from dilute mineral acid.

2. To find the accurate volume of H_2 gas at the given condition of temperature and pressure or to equalize the pressure exerted by moist H_2 gas with the atmospheric pressure.

3. Dalton's law is used to find the partial pressure of dry hydrogen gas collected over water.

Since the total pressure includes both H_2 and H_2O vapour,

Partial pressure of dry $\text{H}_2 = \text{Total pressure} - \text{Aqueous tension (H}_2\text{O vapour pressure)}$

4. Equivalent weight of a metal is the weight of the metal that reacts with or displaces 1.008 g of H_2 , 8 g of O_2 or 35.5 g of Cl_2 .

5. Equivalent weight = $\frac{\text{Molecular weight}}{\text{Basicity}}$

or, E.W. of $\text{H}_3\text{PO}_4 = \frac{3 \times 1 + 31 + 4 \times 16}{3} (\because \text{Basicity} = 3)$

$$= \frac{98}{3} = 32.66 \text{ (no unit)}$$